

Modeling the Dynamics of Alzheimer's Disease

Winthrop University

Landon Loveless Garrett Nix Steven Stokes

Table of Contents

1 Introduction

■ A Brief Synopsis of Alzheimer's Disease

2 Formation of Proposed Model

What Alzheimer's Disease Is

Alzheimer's Disease (AD) is a neurodegenerative disease characterized by many complex interactions that affect over 50 million people worldwide. Many different aspects of the disease are anticipated, but it is not yet known how these factors interact [2].

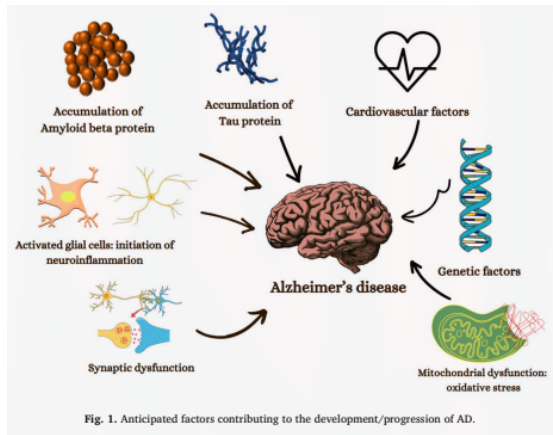


Figure: Various factors of AD from the literature

Hypotheses Galore

Various hypotheses have been proposed in order to understand the underlying mechanisms of AD; where the leading hypothesis for many years has been the *Amyloid Cascade Hypothesis*. However, there has been increasing support in the notion that it is insufficient in giving an effective description and treatment towards AD [1].

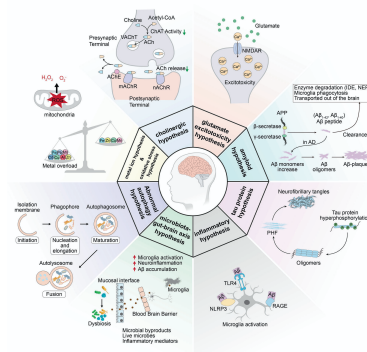


Figure: A visualization of the various hypotheses relating to the mechanisms of AD

Formation of a Newly Proposed Model

In order to understand these factors and subsequently the behavior of AD, many systems of ordinary differential equations (ODEs) have been proposed related to these hypotheses; but, a few are highlighted in regard to creating a newly proposed dynamical system:

- Observing the dynamics between $A\beta$ and Ca^{2+} [3].
- Using causal models observing factors of genetic, non-amyloid dependent, and biomarker cascades [4].

The Proposed Model

Due to the feedback loop created by $A\beta$ and Ca^{2+} , along with the interaction between the hyperphosphorylation of τ -protein caused by $A\beta$, a proposed causal model is described by the following system of ODEs:

$$\frac{dA\beta}{dt} = a_1 - r_1 A\beta + a_2 \frac{Ca}{Ca + \sigma} \quad (1)$$

$$\frac{dCa}{dt} = b_1 - r_2 Ca + b_2 A\beta \quad (2)$$

$$\frac{d\tau}{dt} = c_1 - r_3 \tau + c_2 A\beta + c_3 Ca \quad (3)$$

$$\frac{dN}{dt} = d_1 - k_4 N + d_2 \tau \quad (4)$$

$$\frac{dC}{dt} = e_1 - k_5 C + e_2 NR + e_3 \tau. \quad (5)$$

Discussion of Proposed Model

References

- [1] Kasper P Kepp et al. “The amyloid cascade hypothesis: an updated critical review”. In: *Brain* 146.10 (2023), pp. 3969–3990.
- [2] Mitali Maji and Subhas Khajanchi. “Mathematical models on Alzheimer’s disease and its treatment: A review”. In: *Physics of Life Reviews* 52 (2025), pp. 207–244.
- [3] Swadesh Pal and Roderick Melnik. “Nonequilibrium landscape of amyloid-beta and calcium ions in application to Alzheimer’s disease”. In: *Physical Review E* 111.1 (2025), p. 014418.
- [4] Jeffrey R Petrella et al. “Computational causal modeling of the dynamic biomarker cascade in Alzheimer’s disease”. In: *Computational and mathematical methods in medicine* 2019.1 (2019), p. 6216530.

Acknowledgments

This work was supported primarily by the National Science Foundation EPSCoR Program under NSF Award OIA-2242812.

Closing

Thank you. Questions?